

Integrating Suricata IDS with Wazuh for Network Threat Detection

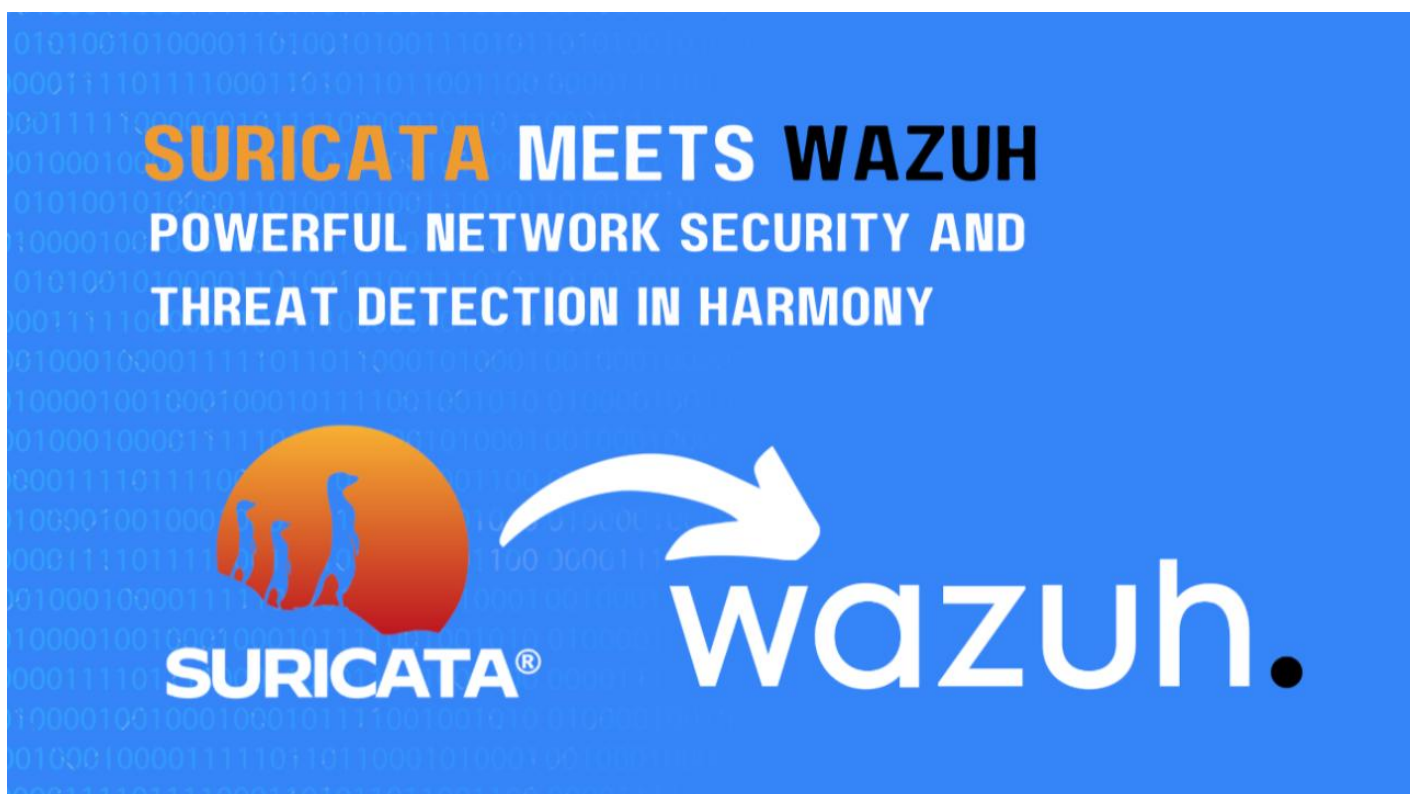
Project Description

This project focuses on deploying **Suricata**, an open-source Network Intrusion Detection System (NIDS), and integrating it with **Wazuh** to achieve centralized security monitoring and visibility of network-based attacks.

The objective is to:

- Detect suspicious and malicious network activity using Suricata
- Forward Suricata alerts to Wazuh
- Visualize and analyze network threats from a single security platform

This project demonstrates how network-level detection complements host-based security monitoring in a SOC environment.



Tools & Technologies Used

- **Suricata** – Network Intrusion Detection System (IDS)
- **Wazuh Manager** – Security monitoring and correlation platform
- **Wazuh Agent** – Log collection from monitored host
- **Kali Linux** – Attacker machine for testing
- **Ubuntu Linux** – Target system with Wazuh Agent
- **Emerging Threats Ruleset** – Detection signatures
- **EVE JSON logs** – Structured Suricata alert output

Why Integrate Suricata with Wazuh?

- Suricata detects **network-based attacks** (scans, exploits, suspicious traffic)
- Wazuh provides:
 - Centralized log management
 - Correlation and alerting
 - Dashboards and historical analysis
- Together they provide **defense-in-depth**:
 - Network visibility (Suricata)
 - Host visibility (Wazuh)

Architecture Overview

Kali Attacker

|
| (Network Traffic / Attacks)

v

Target Machine (Suricata + Wazuh Agent)

|
| Suricata alerts (eve.json)

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Wazuh Manager

|
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Wazuh Dashboard (Alerts & Analysis)

Implementation Steps

Step 1: Install Suricata

Purpose:

Install the Suricata IDS on the target system.

Actions Performed:

- Added Suricata stable repository
- Installed Suricata package
- Verified installation and version

Result:

- Suricata installed successfully
- Version verified using `suricata -V`

```
root@ip-...:/home/ubuntu# sudo add-apt-repository ppa:oisf/suricata-stable
sudo apt-get update
sudo apt-get install suricata -y
```

```
Setting up sse3-support (21build1) ...
Setting up libhiredis1.1.0:amd64 (1.2.0-6ubuntu3) ...
Setting up libhyperscan5 (5.4.2-2) ...
Setting up suricata (1:8.0.3-0ubuntu0) ...
info: Selecting UID from range 100 to 999 ...

info: Selecting GID from range 100 to 999 ...
info: Adding system user `suricata' (UID 109) ...
info: Adding new group `suricata' (GID 115) ...
info: Adding new user `suricata' (UID 109) with group `suricata' ...
info: Not creating home directory `/var/lib/suricata'.
Created symlink /etc/systemd/system/multi-user.target.wants/suricata.service → /usr/lib/systemd/system/suricata.service.
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
```

```
root@ip-...:/home/ubuntu# suricata -V
This is Suricata version 8.0.3 RELEASE
```

Step 2: Download Emerging Threats Rules

Purpose:

Enable Suricata to detect known attack patterns using community rules.

Actions Performed:

- Downloaded Emerging Threats rules
- Extracted rules into /etc/suricata/rules/
- Verified rules directory contents

Result:

- Suricata loaded with a comprehensive detection ruleset

```
root@ip-...:/home/ubuntu# cd /tmp
curl -LO https://rules.emergingthreats.net/open/suricata-6.0.8/emerging.rules.tar.gz
sudo tar -xvzf emerging.rules.tar.gz
sudo mkdir -p /etc/suricata/rules
sudo mv rules/*.rules /etc/suricata/rules/
sudo chmod 644 /etc/suricata/rules/*.rules
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
100 5208k	100 5208k	0 0	59.0M 0	--:--:--	--:--:--	--:--:--	58.4M

```
rules/
rules/BSD-License.txt
rules/LICENSE
rules/botcc.portgrouped.rules
rules/botcc.rules
rules/ciarmy.rules
rules/classification.config
rules/compromised-ips.txt
rules/compromised.rules
rules/drop.rules
rules/dshield.rules
rules/emerging-activex.rules
rules/emerging-adware_pup.rules
rules/emerging-attack_response.rules
rules/emerging-chat.rules
rules/emerging-coinminer.rules
```

Step 3: Configure Suricata

Purpose:

Define network scope, interfaces, and rule loading behavior.

Actions Performed:

- Updated suricata.yaml:
 - Defined HOME_NET
 - Configured network interface (AF_PACKET)
 - Enabled EVE JSON logging
 - Set default rule path and rule files

Result:

- Suricata configured for the target network
- JSON logging enabled for integration

```
## Step 1: Inform Suricata about your network
##

vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[10.0.1.200]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

    #EXTERNAL_NET: "!$HOME_NET"
    EXTERNAL_NET: "any"

    HTTP_SERVERS: "$HOME_NET"
    SMTP_SERVERS: "$HOME_NET"
    SQL_SERVERS: "$HOME_NET"
```

```
# Linux high speed capture support
af-packet:
  - interface: eth0
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets based on flow.
    cluster-id: 99
```

```
default-rule-path: /etc/suricata/rules
```

```
rule-files:
  - "*.rules"
```


Step 4: Restart and Verify Suricata

Purpose:

Ensure Suricata runs correctly with the new configuration.

Actions Performed:

- Restarted Suricata service
- Checked service status
- Verified Suricata running in IDS mode

Result:

- Suricata running successfully
- No configuration errors

```
root@ip-10-0-1-208:/tmp# sudo systemctl restart suricata
sudo systemctl status suricata
● suricata.service - Suricata IDS/IPS/NSM/FW daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-01-26 18:23:58 UTC; 22ms ago
     Docs: man:suricata(8)
           man:suricatasc(8)
           https://suricata.io/documentation/
   Process: 4059 ExecStartPre=/bin/rm -f /run/suricata.pid (code=exited, status=0/SUCCESS)
  Main PID: 4060 (Suricata-Main)
    Tasks: 1 (limit: 4667)
   Memory: 1.8M (peak: 1.8M)
      CPU: 26ms
   CGroup: /system.slice/suricata.service
           └─4060 /usr/bin/suricata --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid --user suricata --group suricata

Jan 26 18:23:58 ip-10-0-1-208 systemd[1]: Starting suricata.service - Suricata IDS/IPS/NSM/FW daemon...
Jan 26 18:23:58 ip-10-0-1-208 systemd[1]: Started suricata.service - Suricata IDS/IPS/NSM/FW daemon.
Jan 26 18:23:58 ip-10-0-1-208 suricata[4060]: i: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode
root@ip-10-0-1-208:/tmp#
```

```
root@ip-10-0-1-208:/tmp# sudo tail -f /var/log/suricata/eve.json
{"timestamp":"2026-01-26T18:24:27.828659+0000","flow_id":1025790668248392,"in_iface":"enX0","event_type":"alert","src_ip":"198.235.24.76","src_port":56313,"dest_ip":"10.0.1.208","dest_port":4433,"proto":"TCP","ip_v":4,"pkt_src":"wire/packet","metadata":{"flowbits":["ET.Evil","ET.DshieldIP"]},"alert":{"action":"allowed","gid":1,"signature_id":2402000,"rev":7628,"signature":"ET DROP Dshield Block Listed Source group 1","category":"Misc Attack","severity":2,"metadata":{"affected_product":["Any"],"attack_target":["Any"],"created_at":["2010_12_30"],"deployment":["Perimeter"],"signature_severity":["Major"],"tag":["Dshield"],"updated_at":["2026_01_23"]},"direction":"to_server","flow":{"pkts_to_server":1,"pkts_to_client":0,"bytes_to_server":60,"bytes_to_client":0,"start":"2026-01-26T18:24:27.828659+0000","src_ip":"198.235.24.76","dest_ip":"10.0.1.208","src_port":56313,"dest_port":4433}}
{"timestamp":"2026-01-26T18:24:33.723280+0000","event_type":"stats","stats":{"uptime":35,"capture":{"kernel_packets":15,"kernel_drops":0,"errors":0,"afpacket":{"busy_loop_avg":0,"polls":122,"poll_signal":0,"poll_timeout":116,"poll_data":6,"poll_errors":0,"send_errors":0},"decoder":{"pkts":15,"bytes":1884,"invalid":0,"ipv4":15,"ipv6":0,"ethernet":15,"arp":0,"unknown_ethertype":0,"chdlc":0,"raw":0,"null":0,"sll":0,"sll2":0,"tcp":13,"udp":2,"sctp":0,"esp":0,"icmpv4":0,"icmpv6":0,"ppp":0,"pppoe":0,"geneve":0,"gre":0,"vlan":0,"vlan_qinq":0,"vlan_qinqing":0,"vxlan":0,"vntag":0,"ieee8021ah":0,"teredo":0,"ipv4_in_ipv4":0,"ipv6_in_ipv4":0,"ipv4_in_ipv6":0,"ipv6_in_ipv6":0,"mpls":0,"avg_pkt_size":125,"max_pkt_size":256,"max_mac_addr_src":0,"max_mac_addr_dst":0,"erspan":0,"nsh":0,"event":{"afpacket":{"trunc_pkt":0},"ipv4":{"pkt_too_small":0,"hlen_too_small":0,"iplen_smaller_than_hlen":0,"trunc_pkt":0,"opt_invalid":0,"opt_invalid_len":0,"opt_malformed":0,"opt_pad_required":0,"opt_eol_required":0,"opt_duplicate":0,"opt_unknown":0,"wrong_ip_version":0,"icmpv6":0,"frag_pkt_too_large":0,"frag_overlap":0,"frag_ignored":0},"icmpv4":{"pkt_too_small":0,"unknown_type":0,"unknown_code":0},"ipv4_trunc_pkt":0,"ipv4_unknown_ver":0}}}}
```

Step 5: Configure Wazuh Agent to Read Suricata Logs

Purpose:

Allow Wazuh to ingest Suricata alerts.

Actions Performed:

- Added Suricata EVE JSON log path to Wazuh agent configuration:

/var/log/suricata/eve.json

- Set log format to JSON
- Restarted Wazuh Agent

Result:

- Wazuh agent started collecting Suricata logs

```
<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>
```

```
root@ip-10-10-10-10: /tmp# sudo nano /var/ossec/etc/ossec.conf
root@ip-10-10-10-10: /tmp# sudo systemctl restart wazuh-agent
sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-01-26 18:25:54 UTC; 32ms ago
 Process: 4179 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 27 (limit: 4667)
   Memory: 18.0M (peak: 20.1M)
      CPU: 6.789s
   CGroup: /system.slice/wazuh-agent.service
           └─4203 /var/ossec/bin/wazuh-execd
             └─4213 /var/ossec/bin/wazuh-agentd
               └─4227 /var/ossec/bin/wazuh-syscheckd
                 └─4240 /var/ossec/bin/wazuh-logcollector
                   └─4265 /var/ossec/bin/wazuh-modulesd

Jan 26 18:25:48 ip-10-10-10-10 systemd[1]: Starting wazuh-agent.service - Wazuh agent...
Jan 26 18:25:48 ip-10-10-10-10 env[4179]: Starting Wazuh v4.14.1...
Jan 26 18:25:49 ip-10-10-10-10 env[4179]: Started wazuh-execd...
Jan 26 18:25:50 ip-10-10-10-10 env[4179]: Started wazuh-agentd...
Jan 26 18:25:51 ip-10-10-10-10 env[4179]: Started wazuh-syscheckd...
Jan 26 18:25:52 ip-10-10-10-10 env[4179]: Started wazuh-logcollector...
Jan 26 18:25:52 ip-10-10-10-10 env[4179]: Started wazuh-modulesd...
Jan 26 18:25:54 ip-10-10-10-10 env[4179]: Completed.
Jan 26 18:25:54 ip-10-10-10-10 systemd[1]: Started wazuh-agent.service - Wazuh agent.
```

Step 6: Attack Simulation & Testing

Purpose:

Validate detection and log ingestion.

Actions Performed:

- From Kali attacker machine:
 - ICMP ping test
 - Nmap scan against target
- Observed generated Suricata alerts

Result:

- Suricata detected:
 - Port scans
 - Suspicious inbound traffic
- Alerts written to eve.json

```

(kali㉿ kali)-[~]
$ ping -c 3 10.0.1.200
PING 10.0.1.200 (10.0.1.200) 56(84) bytes of data.
64 bytes from 10.0.1.200: icmp_seq=1 ttl=64 time=0.849 ms
64 bytes from 10.0.1.200: icmp_seq=2 ttl=64 time=0.992 ms
64 bytes from 10.0.1.200: icmp_seq=3 ttl=64 time=0.863 ms

--- 10.0.1.200 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2019ms
rtt min/avg/max/mdev = 0.849/0.901/0.992/0.064 ms

(kali㉿ kali)-[~]
$ nmap -sS 10.0.1.200
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-26 18:45 +0000
Nmap scan report for 10.0.1.200
Host is up (0.00045s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 0A:FF:EB:3F:77:93 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.73 seconds

```

Step 7: Alert Visibility in Wazuh

Purpose:

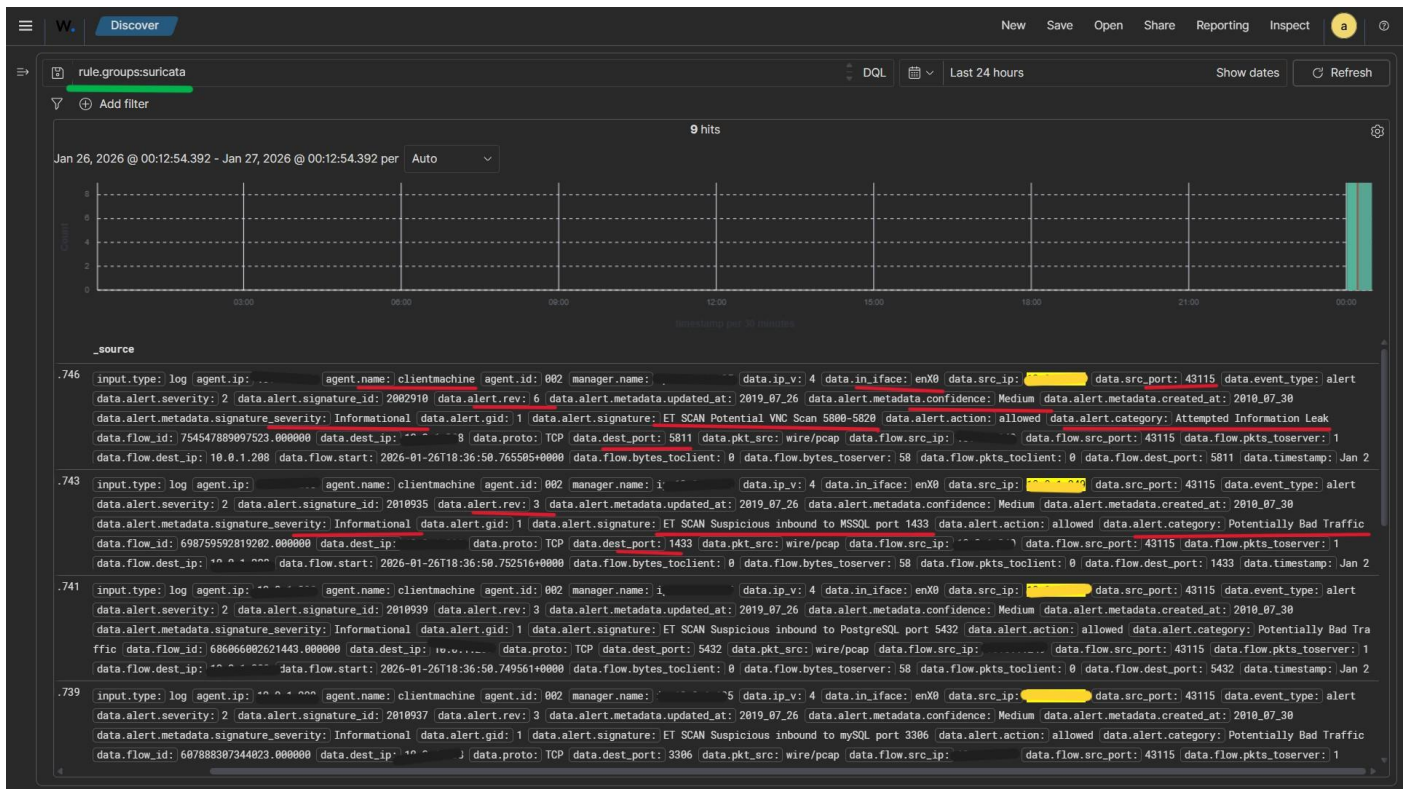
Confirm successful integration and visualization.

Actions Performed:

- Viewed alerts in Wazuh Dashboard
- Analyzed fields:
 - Source IP
 - Destination IP
 - Alert signature
 - Severity
 - Protocol and ports

Result:

- Suricata alerts visible in Wazuh Discover
- Clear mapping of network attack details



t	_index	wazuh-alerts-4.x-2026.01.26
t	agent.id	002
t	agent.ip	
t	agent.name	clientmachine
t	data.alert.action	allowed
t	data.alert.category	Attempted Information Leak
t	data.alert.gid	1
t	data.alert.metadata.confidence	Medium
t	data.alert.metadata.created_at	2010_07_30
t	data.alert.metadata.signature_severity	Informational
t	data.alert.metadata.updated_at	2019_07_26
t	data.alert.rev	6
t	data.alert.severity	2
t	data.alert.signature	ET SCAN Potential VNC Scan 5800-5820
t	data.alert.signature_id	2002910
t	data.dest_ip	
t	data.dest_port	5811
t	data.direction	to_server
t	data.event_type	alert
t	data.in_iface	enx0
t	data.ip_v	4
t	data.pkt_src	wire/pcap
t	data.proto	TCP
t	data.src_ip	
t	data.src_port	43115
📅	data.timestamp	Jan 27, 2026 @ 00:06:50.765
t	decoder.name	json
t	id	1769452611.2586670
t	input.type	log
t	location	/var/log/suricata/eve.json
t	manager.name	
t	rule.description	Suricata: Alert - ET SCAN Potential VNC Scan 5800-5820
#	rule.firedtimes	113
t	rule.groups	ids, suricata
t	rule.id	86601
#	rule.level	3
🕒	rule.mail	false
📅	timestamp	Jan 27, 2026 @ 00:06:51.746

t _index	wazuh-alerts-4.x-2026.01.26
t agent.id	002
t agent.ip	[REDACTED]
t agent.name	clientmachine
t data.alert.action	allowed
t data.alert.category	Potentially Bad Traffic
t data.alert.gid	1
t data.alert.metadata.confidence	Medium
t data.alert.metadata.created_at	2010_07_30
t data.alert.metadata.signature_severity	Informational
t data.alert.metadata.updated_at	2019_07_26
t data.alert.rev	3
t data.alert.severity	2
t data.alert.signature	ET SCAN Suspicious inbound to MSSQL port 1433
t data.alert.signature_id	2010935
t data.dest_ip	[REDACTED]
t data.dest_port	1433
t data.direction	to_server

t data.flow_id	698759592819202.000000
t data.in_iface	enX0
t data.ip_v	4
t data.pkt_src	wire/pcap
t data.proto	TCP
t data.src_ip	[REDACTED]
t data.src_port	43115
📅 data.timestamp	Jan 27, 2026 @ 00:06:50.752
t decoder.name	json
t id	1769452611.2584852
t input.type	log
t location	/var/log/suricata/eve.json
t manager.name	i [REDACTED]
t rule.description	Suricata: Alert - ET SCAN Suspicious inbound to MSSQL port 1433
# rule.firedtimes	112
t rule.groups	ids, suricata
t rule.id	86601
# rule.level	3
🔔 rule.mail	false

Project Outcome

- Suricata successfully detects network attacks
- Wazuh ingests and indexes Suricata alerts
- SOC-style visibility achieved for network threats
- End-to-end IDS → SIEM integration validated

Skills Gained

- Network Intrusion Detection (IDS)
- Suricata deployment and tuning
- Rule-based threat detection
- Log forwarding and SIEM integration
- SOC-style alert analysis
- Attack simulation and validation

Conclusion

This project demonstrates a practical implementation of **network-based threat detection using Suricata integrated with Wazuh**. By combining Suricata's deep packet inspection with Wazuh's centralized monitoring capabilities, we achieve enhanced visibility into network attacks and suspicious activity.

This setup forms a strong foundation for advanced security operations and can be extended further with custom rules, active response, and automated incident handling in future projects.
